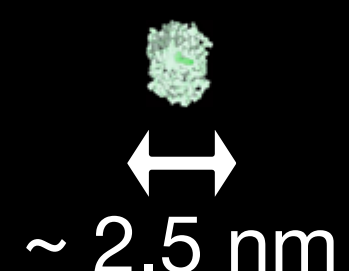


The Diffraction Limit

Fluorescent proteins frequently used in biological research are small

Green Fluorescent Protein (GFP)

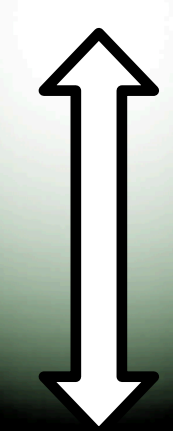


Even smaller than the wavelength of light they emit (~ 500 nm)

Interference of light results in a point-spread function (PSF) like the Airy disk

Structures smaller than the critical distance cannot be resolved by traditional light microscopes

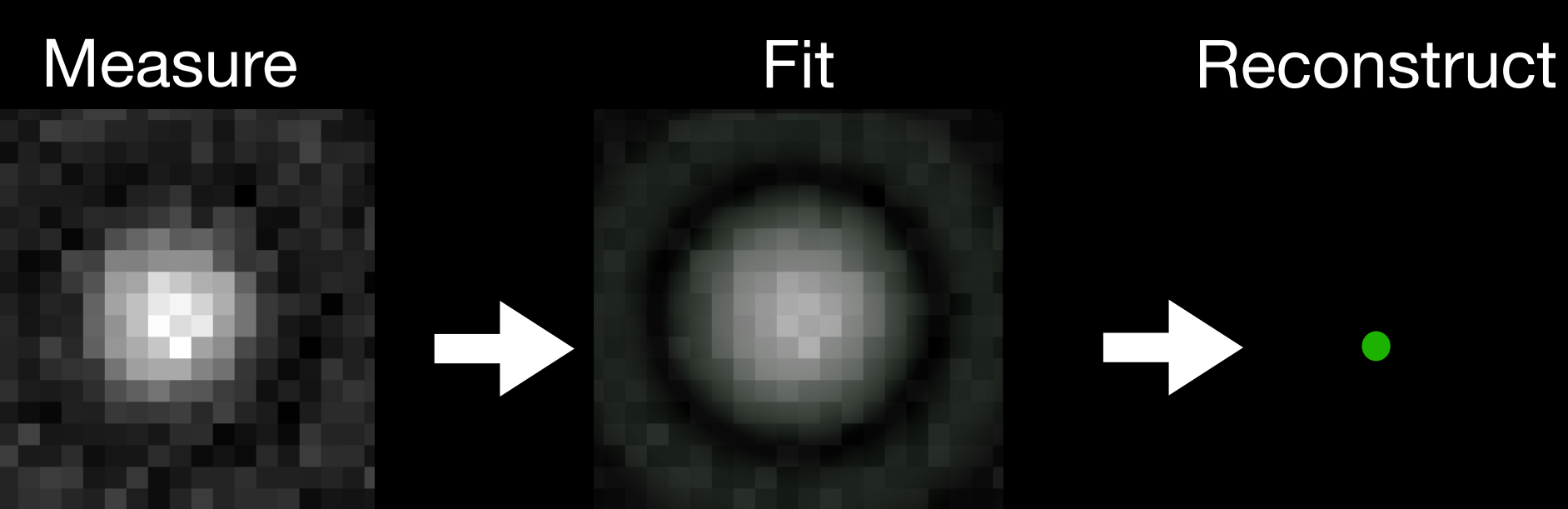
$d \approx 200$ nm



Rayleigh: $d \geq 0.61 \frac{\lambda}{NA}$ Abbe: $d \geq 0.5 \frac{\lambda}{NA}$

Super-Resolution Microscopy

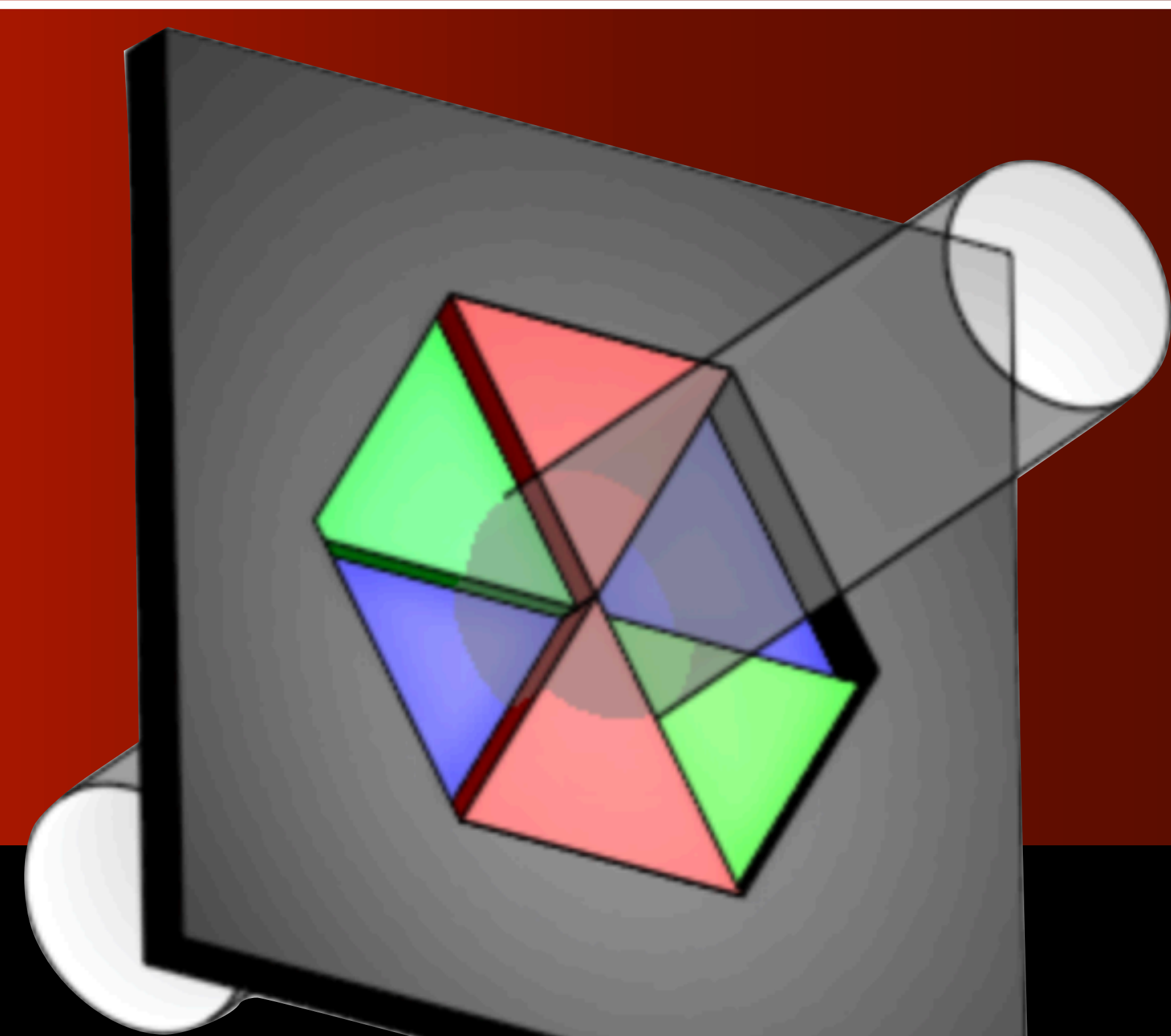
Maximum-likelihood estimation of structure beyond the diffraction limit is possible



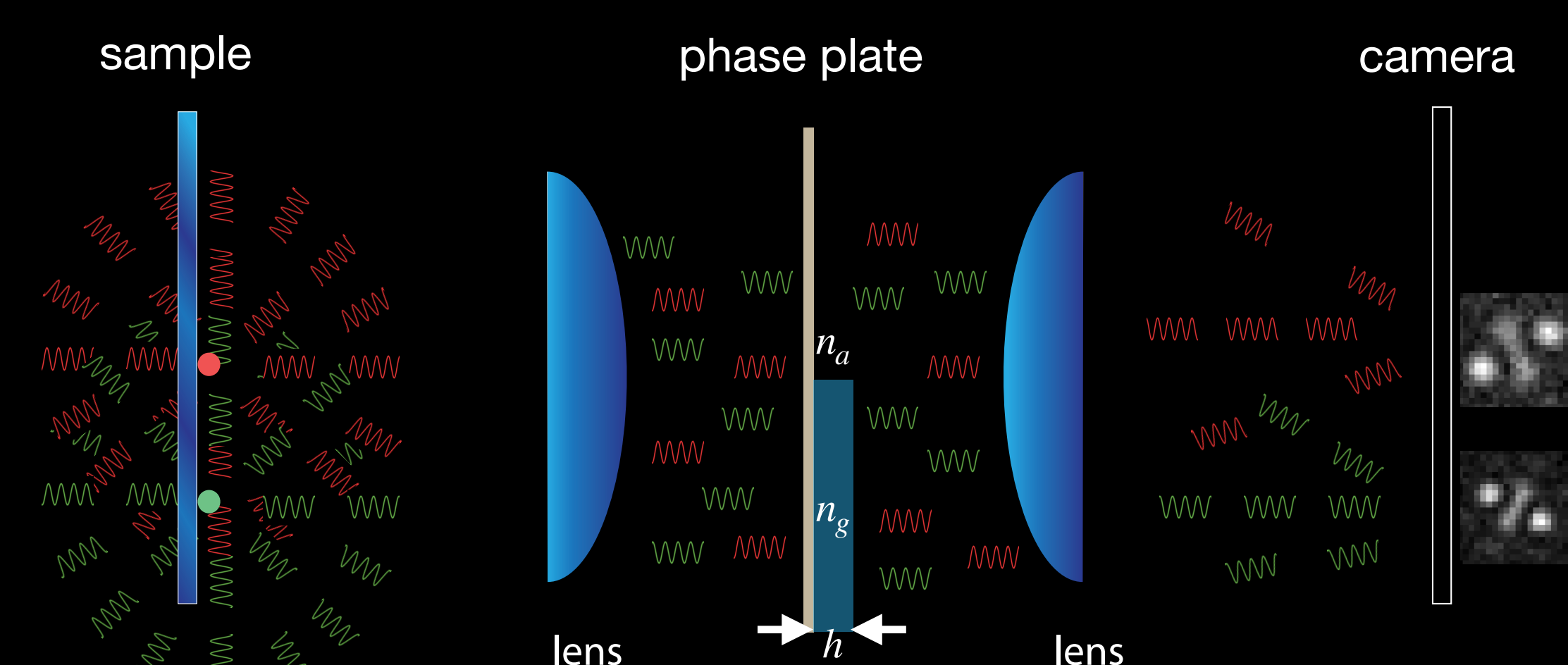
Single-molecule localization methods can achieve estimation precision up to the *Cramer-Rao Bound* (~10 nm), derived from the Fisher Information

$$CRB_{\theta_i}(\vec{\theta}) \leq \sqrt{F_{ii}^{-1}(\vec{\theta})}; \quad \vec{\theta} = [x_0, y_0, z_p, \lambda, \dots]$$

STORM, PALM, PAINT, and many other methods can visualize nanometer-scale biology



PSF Engineering with a Glass Phase Plate



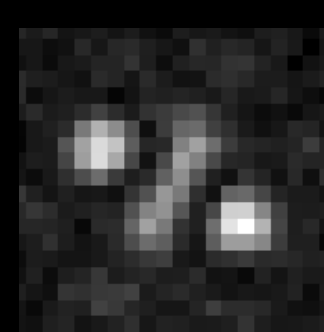
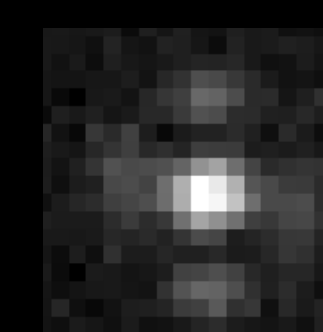
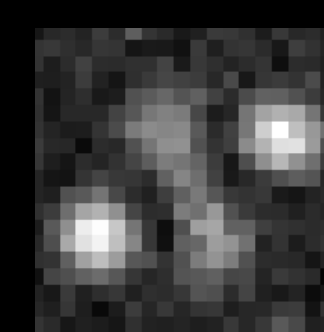
The phase plate imparts a wavelength-dependent phase shift to the emission wavefront to make color-coded PSFs

$$\Delta\phi(\lambda) = \frac{2\pi}{\lambda}(n_g - n_a)h$$

Chernoff Information

The best way to classify an observation

$\nu =$  as either

$\nu_1 =$  or $\nu_2 =$  or $\nu_3 =$ 

is to compare the Chernoff Information (or Chernoff Bound) it has with each known class. This quantity tightly bounds the probability of a misclassification error (Bayes' error)

$$C(\beta_i) = \sum_{\nu_i} \nu_i \frac{(\beta_i - 1)(\log(\frac{\beta_i - 1}{\log \beta_i}) - 1) + \log \beta_i}{\log \beta_i}; \quad \beta_i = \frac{\nu}{\nu_i}$$

The class that provides the tightest bound is the best classification decision we can make

If $C(\beta_1) < C(\beta_2)$ and $C(\beta_1) < C(\beta_3)$, then $\nu \rightarrow \nu_1$

More Chernoff Information between classes yields more distinguishable PSF patterns

A Hexagonal Phase Plate for Optimal 3-Color Super-Resolution Microscopy

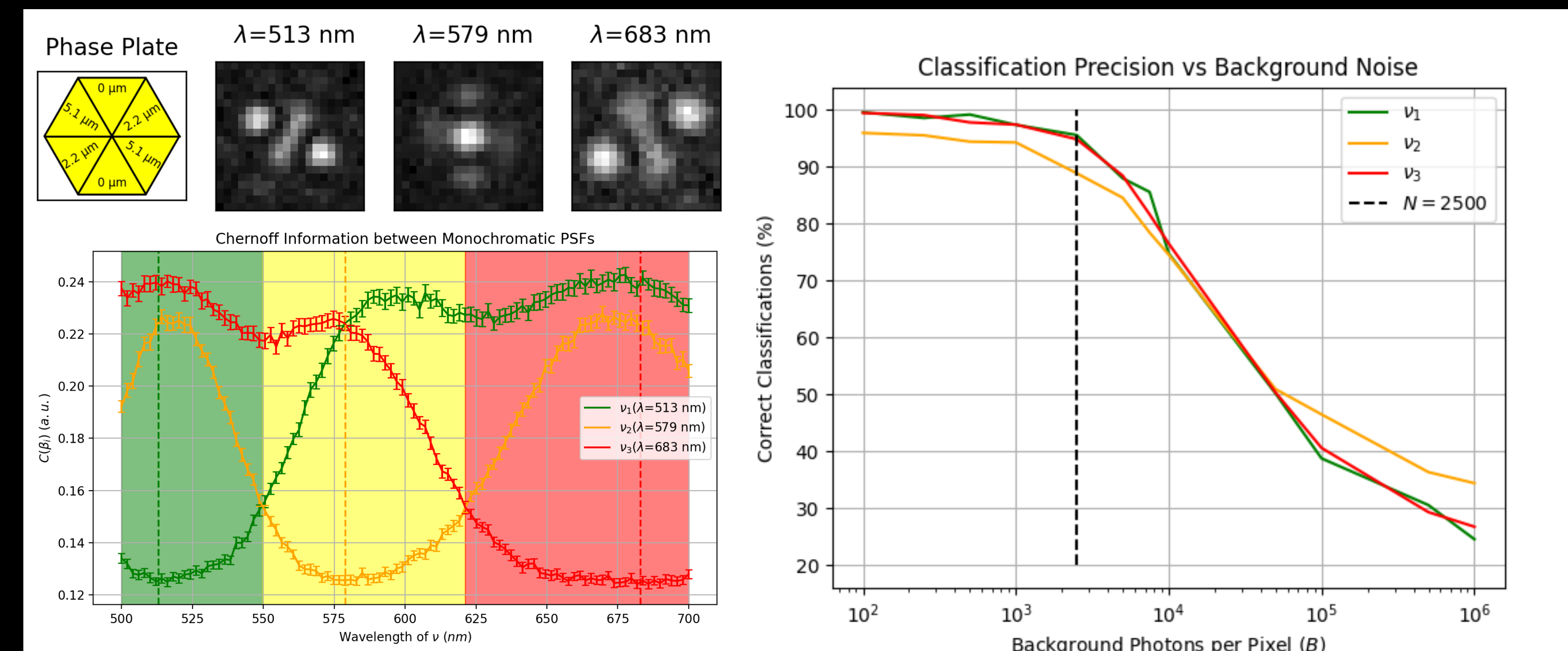
Matthew Hogan, Dept. of Physics & Astronomy

Jordan Gerton, Dept. of Physics & Astronomy

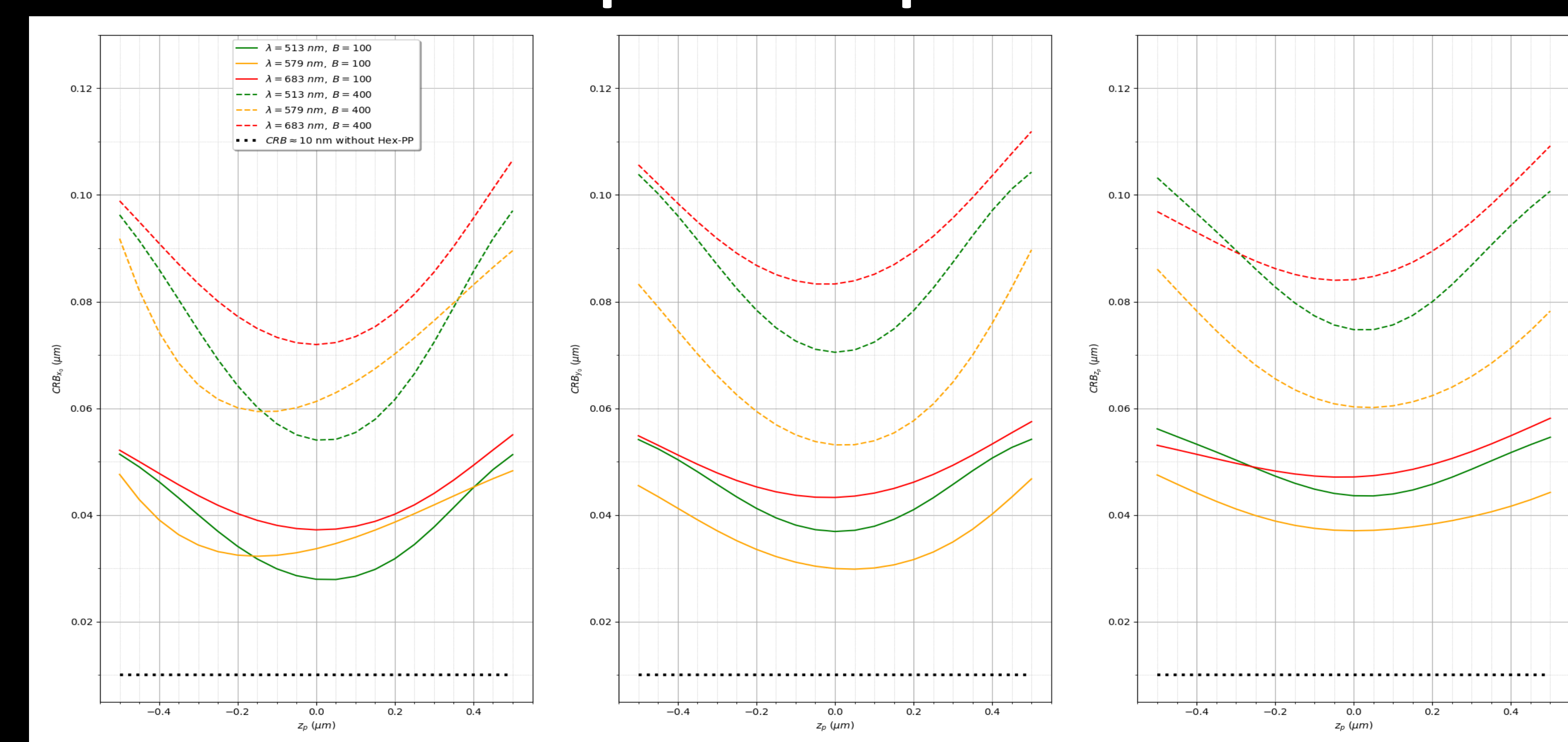
Erik Jorgensen, School of Biological Sciences

In collaboration with Brian Baker and Utah NanoFab

Improved Fluorophore Classification...

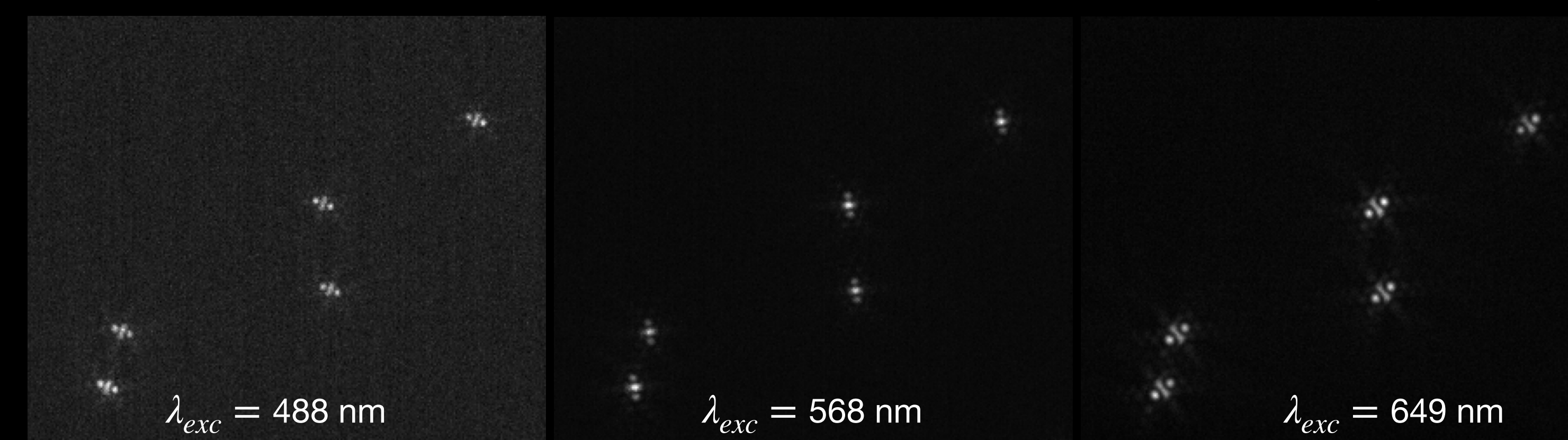


... at the Expense of Spatial Precision



The results above describe the best-case precisions for simulated Hex-PSFs of $N = 2500$ photons

Diffraction-Limited Fluorescent Microspheres



Sequential laser excitations of a ~400 μm^2 field of view of 0.1 μm Tetraspeck™ microspheres. Each microsphere has green (513 nm), yellow (579 nm), and red (683 nm) fluorescent molecules attached

References

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